Trigonometric Substitution

7.3



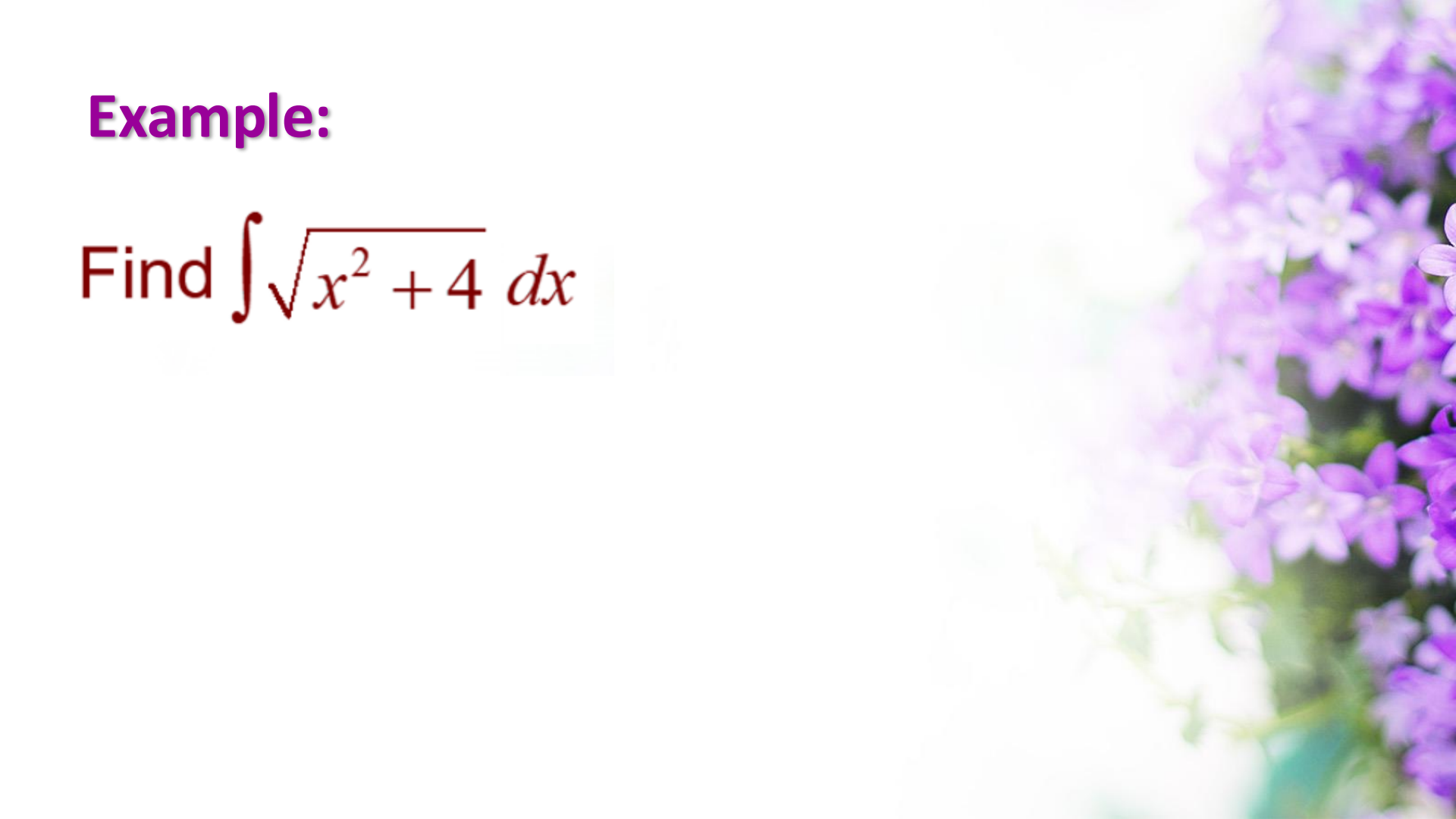
Our goals:

- Solve some integration that is involving roots using various types of trigonometric substitutions.

- Finding the area of some shapes like circle and Ellipse.

Example:

Find $\int \sqrt{x^2+4}\, dx$



Integration techniques:

The Substitution Rule

$$\int f[g(x)]g'(x)dx \longrightarrow \int f(u)du$$

Integration By Parts

$$\int u\,dv = uv - \int v\,du$$



Integration by Trigonometric Substitution

$$\sqrt{a^2 - x^2} \qquad \sqrt{a^2 + x^2} \qquad \sqrt{x^2 - a^2}$$

applicable when one of these expressions is in the integrand

(a is any positive number)

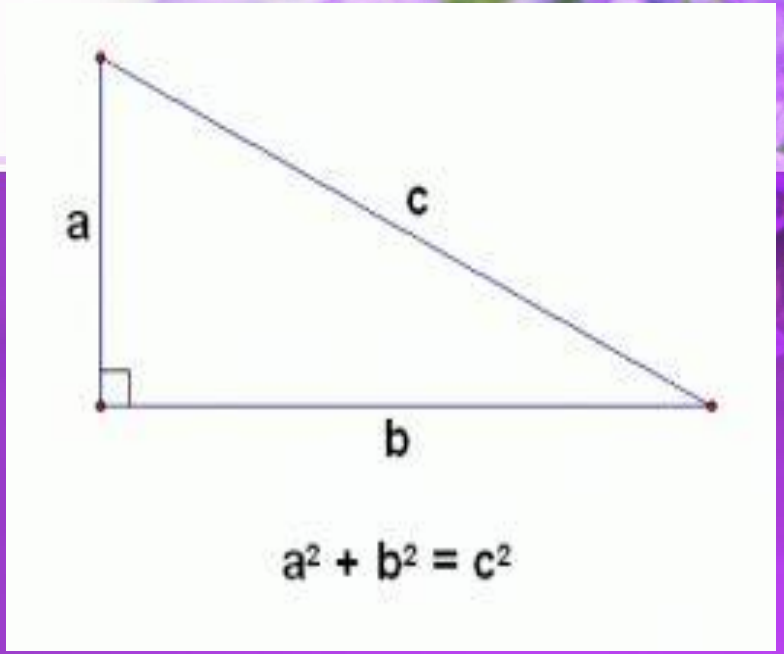


Can we use integration by substitution to solve such type of problem?

$$\sqrt{a^2 - x^2} \qquad \sqrt{a^2 + x^2} \qquad \sqrt{x^2 - a^2}$$

we will need the following formulas:

$$\sqrt{x^2} = |x|$$



$$cos^2\theta + sin^2\theta = 1$$

$$cos^2\theta = 1 - sin^2\theta$$

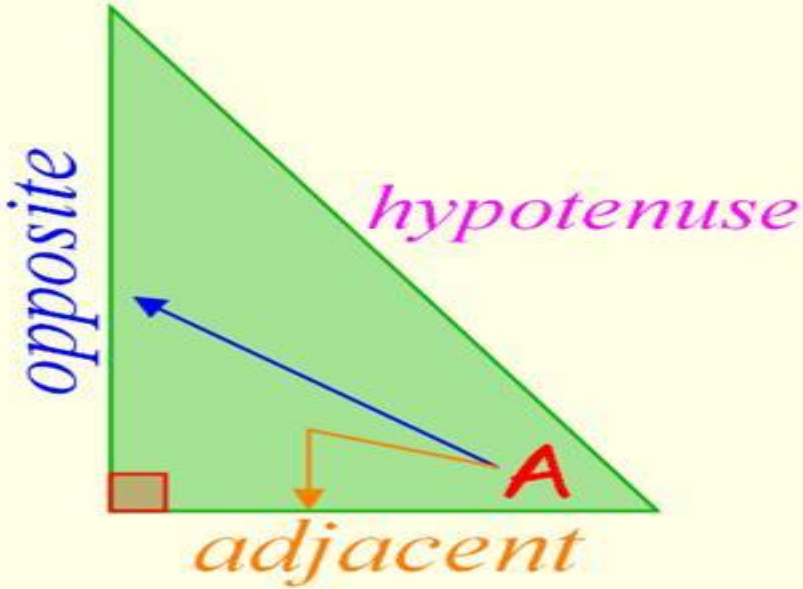
$$1 + tan^2\theta = sec^2\theta$$

$$tan^2\theta = sec^2\theta - 1$$

$$\sin A = \frac{opposite}{hypotenuse}$$

$$\cos A = \frac{adjacent}{hypotenuse}$$

$$\tan A = \frac{opposite}{adjacent}$$



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Integration by Trigonometric Substitution

$\sqrt{a^2 - x^2}$	$\sqrt{a^2 + x^2}$	$\sqrt{x^2 - a^2}$
$x = a\sin\theta$	$x = a\tan\theta$	$x = a\sec\theta$

TABLE OF TRIGONOMETRIC SUBSTITUTIONS

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin\theta, \quad -\frac{\pi}{2} \leqslant \theta \leqslant \frac{\pi}{2}$	$1 - \sin^2\theta = \cos^2\theta$
$\sqrt{a^2 + x^2}$	$x = a \tan\theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$1 + \tan^2\theta = \sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a \sec\theta, \quad 0 \leqslant \theta < \frac{\pi}{2}$ or $\pi \leqslant \theta < \frac{3\pi}{2}$	$\sec^2\theta - 1 = \tan^2\theta$

Integration by Trigonometric Substitution

$\sqrt{a^2 - x^2} \rightarrow a\cos\theta$	$\sqrt{a^2 + x^2} \rightarrow a\sec\theta$	$\sqrt{x^2 - a^2} \rightarrow a\tan\theta$
$x = a\sin\theta$	$x = a\tan\theta$	$x = a\sec\theta$

Example:

Evaluate $\int \frac{\sqrt{9-x^2}}{x^2}\,dx$

Remember that:

$\sqrt{a^2 - x^2}$	$\sqrt{a^2 + x^2}$	$\sqrt{x^2 - a^2}$
$x = a\sin\theta$	$x = a\tan\theta$	$x = a\sec\theta$

$$\int \sin u \, du = -\cos u + C$$

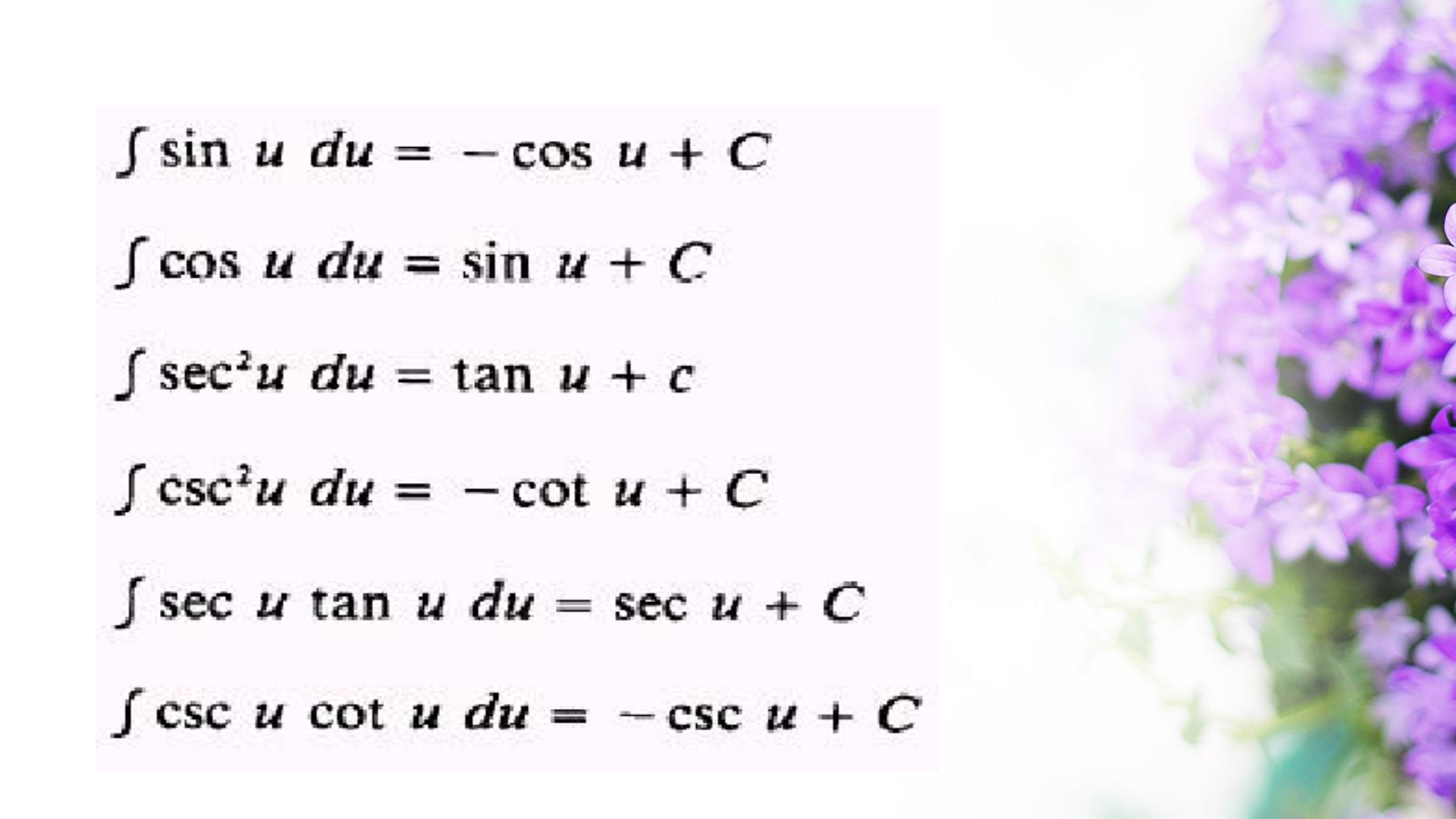
$$\int \cos u \, du = \sin u + C$$

$$\int \sec^2 u \, du = \tan u + c$$

$$\int \csc^2 u \, du = -\cot u + C$$

$$\int \sec u \tan u \, du = \sec u + C$$

$$\int \csc u \cot u \, du = -\csc u + C$$



Summary:

1) Recognize one of the expressions in the integrand and make the appropriate substitution.

2) Manipulate dx to get something with $d\theta$ and then integrate the resulting expression.

3) Write the answer in terms of x.

Example:

Find $\int \frac{x}{\sqrt{x^2+4}}\,dx$

Remember that:

$\sqrt{a^2 - x^2}$	$\sqrt{a^2 + x^2}$	$\sqrt{x^2 - a^2}$
$x = a\sin\theta$	$x = a\tan\theta$	$x = a\sec\theta$

Remark:

Last example illustrates the fact that, even when trigonometric substitutions are possible, they may not give the easiest solution.
You should look for a simpler method first.

Example:

Find $\int \frac{1}{x^2\sqrt{x^2+4}}\,dx$

Remember that:

$\sqrt{a^2 - x^2}$	$\sqrt{a^2 + x^2}$	$\sqrt{x^2 - a^2}$
$x = a\sin\theta$	$x = a\tan\theta$	$x = a\sec\theta$

Example:

Evaluate $\int \frac{dx}{\sqrt{x^2 - 16}}$

Remember that:

$\sqrt{a^2 - x^2}$	$\sqrt{a^2 + x^2}$	$\sqrt{x^2 - a^2}$
$x = a\sin\theta$	$x = a\tan\theta$	$x = a\sec\theta$

Applications:

An Architect Engineer uses integration in determining the amount of the necessary materials to construct curved shape constructions (e.g. dome over a sports arena)

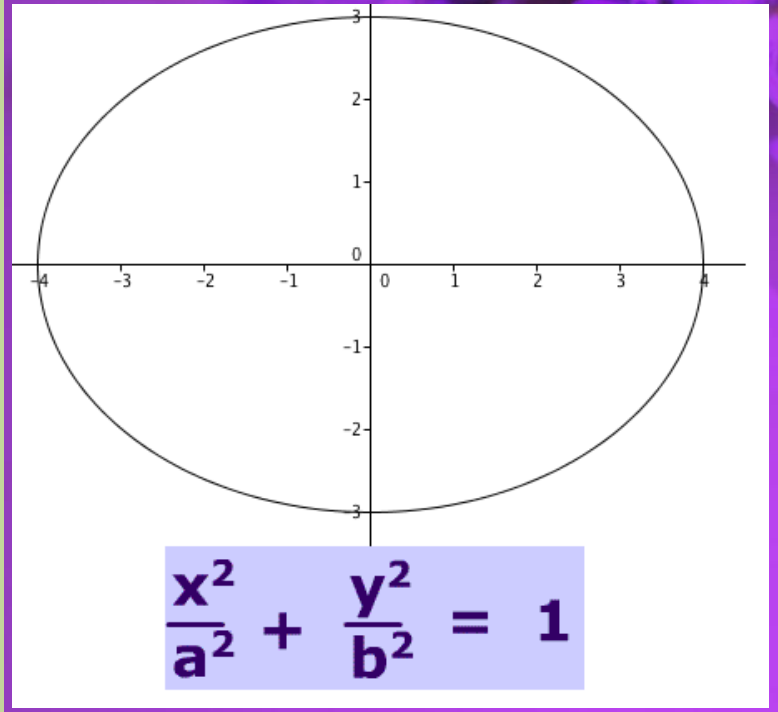
For Example,

The dome of the Taj Mahal in India, built between 1632-1653 by Shah Jahan, involves complex integral calculus to determine the exact shape of the dome.



Remark:

- We can find the area of Ellipse and more complicated integral using the trigonometric substitution



Example: Find

$$\int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2+9)^{3/2}}\,dx$$

Example: Find

$$\int \frac{x}{\sqrt{3-2x-x^2}}\,dx$$